

Exercise 5

Number Theory: Finding the Nth Prime Number

Introduction:

A prime number is a natural number greater than one that has no positive divisors other than one and itself. Prime numbers form the foundation of number theory and appear extensively in cryptography, hashing algorithms, and computational mathematics. Determining whether a given integer is prime is a fundamental algorithmic problem.

A number n is prime if no integer from 2 up to the square root of n divides it evenly. The modulo operator is used to test divisibility:

```
if (n % divisor == 0) // n is divisible by divisor
```

By iterating over the natural numbers and counting those that satisfy the primality condition, the n th prime number in the sequence can be identified. The first few prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29.

Learning Outcomes:

- Implement a function that determines whether a given integer is a prime number.
- Compute and display the n th prime number by applying the primality test across the natural numbers.

Problem Description:

```
/* Determines if integer n is a prime number.
   Returns true if n is prime, false otherwise. */
bool isPrime(int n);

/* Computes and returns the nth prime number in the sequence of prime numbers,
   where n is a positive integer supplied by the caller.
   Uses the function isPrime. */
int getNthPrime(int n);

/* Prints the first count prime numbers, each on a separate line.
   Uses the function isPrime. */
void printPrimes(int count);

/* Reads a positive integer n from the user, displays the nth prime number,
   and prints the first n prime numbers.
   Uses the functions getNthPrime and printPrimes. */
void runPrimeFinder();
```

Call all the functions created in function `main()`.